

Flask Complete Notes (Chapter 1–10)

Chapter 1: Flask Basics

- 1 Flask = Python web framework.
- 2 Used to build backend for web apps.
- 3 Flow: User → Request → Flask → Function → Response.
- 4 Key: app, route, function, return.

Chapter 2: Routing

- 1 Routing maps URL to function.
- 2 Dynamic URLs: /user/
- 3 Types: string, int, float
- 4 Used in real apps like profiles, products.

Chapter 3: Templates (Jinja2)

- 1 HTML files stored in templates folder.
- 2 `render_template()` sends data.
- 3 Jinja2 syntax: `{{ variable }}`
- 4 Used to build UI.

Chapter 4: Forms

- 1 Used to take input from user.
- 2 GET vs POST (POST is secure).
- 3 `request.form` used to get data.
- 4 Flow: Input → Backend → Response.

Chapter 5: Advanced Templates

- 1 Loops: `{% for %}`
- 2 Conditions: `{% if %}`
- 3 Template inheritance using `base.html`.
- 4 Block concept for reusable layout.

Chapter 6: Static Files

- 1 CSS, JS, Images stored in static folder.
- 2 Use `url_for('static', filename='...')`
- 3 Adds styling and interactivity.

Chapter 7: Redirects

- 1 `redirect()` moves user to another route.
- 2 `url_for()` generates URL safely.
- 3 Used after login, form submission.

Chapter 8: Flash Messages

- 1 `flash()` shows temporary messages.
- 2 Requires `secret_key`.
- 3 Used for alerts like success/error.

Chapter 9: Sessions

- 1 Session stores user data.
- 2 `session['user'] = name`
- 3 `session.pop()` for logout.
- 4 Used for login systems.

Chapter 10: Database (SQLite)

- 1 Stores data permanently.
- 2 Operations: CREATE, INSERT, SELECT.
- 3 Use `sqlite3` in Python.
- 4 Used in real apps like login, products.